AN EXPLICIT FORMULA FOR THE LODAY ASSEMBLY

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ABSTRACT. We give an explicit description of the Loday assembly map on homotopy groups when restricted to a certain subgroup coming from the Atiyah-Hirzebruch spectral sequence. This proves and generalises a formula about the Loday assembly map on the first homotopy group written down in the survey article by Lück and Reich. Furthermore, we use this new result to prove the Loday assembly for the integral group ring of cyclic group of order 2 is a bijection on the second homotopy group, and is an injection when one considers n-fold direct sum of cyclic group of order 2.

CONTENTS

1. INTRODUCTION

The general consensus is that computing homotopy groups is difficult. Hence, investigating algebraic K-theory of rings is not an easy task. This is why assembly map comes into play—to approximate homotopy theory with a generalised homology theory. In his dissertation [Lod76], Loday wrote down the first example of a K-theoretic assembly map

$$\alpha_{\text{Loday}} : BG_+ \wedge \mathbb{K}_R^{\text{GW}} \to \mathbb{K}_{R[G]}^{\text{GW}}$$

(see [Lod76, Proposition 4.1.1 on page 356] or Definition 3.1 below). Here, $\mathbb{K}_R^{\text{GW}}$ is the non-connective Gersten-Wagoner algebraic K-theory spectrum of the ring R defined below, and the domain represents the generalised homology theory of the classifying space BG of the group G with coefficients in the spectrum $\mathbb{K}_R^{\mathrm{GW}}$. Furthermore, Loday related the cokernel of the induced map on homotopy groups with the classical Whitehead groups defined by J. H. C. Whitehead in [Whi50] and Hatcher-Wagoner in [HW73] when $R = \mathbb{Z}$. Motivated by the connection between the classical Novikov Conjecture and the rational injectivity of the L-theoretic assembly map, Hsiang cast the following conjecture in his ICM address:

Conjecture 1.1 (K-theoretic Novikov Conjecture, [Hsi84])
When R is regular, and G is torsion-free, the map

$$\pi_i(\alpha_{\mathrm{Loday}}) \otimes \mathrm{id}_{\mathbb{Q}} : \pi_i\left(BG_+ \wedge \mathbb{K}_R^{\mathrm{GW}}\right) \otimes \mathbb{Q} \to K_i(R[G]) \otimes \mathbb{Q} \tag{1.0.1}$$

is injective for all i.

Attempts on answering Conjecture 1.1 often leads to the creation of new mathematics, most notably the Topological Cyclic Homology, which allowed Bökstedt-Hsiang-Madsen [BHM93, Theorem 9.13 on page 535] to prove Conjecture 1.1 when G is homologically finite. A stronger statement is also possible:

Conjecture 1.2 (Classical Farrell-Jones Conjecture)
When R is regular and G is torsion-free, the map

$$\pi_i(\alpha_{\mathrm{Loday}}) : \pi_i(BG_+ \wedge \mathbb{K}_R^{\mathrm{GW}}) \to K_i(R[G]) \tag{1.0.2}$$

is an isomorphism for all i.

This conjecture can be understood as a conceptual approach to computing algebraic K-theory of a group ring via a homology theory, but it has important implications for a range of topics, notably the Borel Conjecture concerning the topological rigidity of closed spherical manifolds, and the Kaplansky Conjecture about the idempotents in a group ring.

One might ask whether torsion-free is a necessary condition in Conjecture 1.2. As stated before, the cokernel of

$$\pi_i(\alpha_{\mathrm{Loday}}) : \pi_i(BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\mathrm{GW}}) \to K_i(\mathbb{Z}[G])$$

can be identified with the Whitehead groups $\mathrm{Wh}_i(G)$. So, non-vanishing $\mathrm{Wh}_i(G)$ implies the non-surjectivity of $\pi_i(\alpha_{\mathrm{Loday}})$. This point is known, for example, when $G = C_p$ is the cyclic group of prime order p [Coh73, 11.5 on page 45] for $i = 1$. What happens with non-injectivity?

Question 1.3
For a regular ring R, is there a group G with torsion that makes $\pi_i(\alpha_{\mathrm{Loday}})$ not injective for some i?

Ullmann-Wu gives an example (the Klein four-group) where non-injectivity can occur when R is a finite field [UW17]. In their case, non-injectivity occurs at the second homotopy group. However, it is currently not known whether such an example exists $R = \mathbb{Z}$.

When $R = \mathbb{Z}$, Question 1.3 is known to be negative for $i = 0, 1$. See [LR05, Lemma 2 on page 709] for example. This paper contributes to these problems by giving an explicit description of $\pi_i(\alpha_{\mathrm{Loday}})$ when restricted onto a subgroup of $\pi_i(BG_+ \wedge \mathbb{K}_R^{\mathrm{GW}})$ coming from the Atiyah-Hirzebruch spectral sequence (see Theorem 3.8), and to Conjecture 1.3 by investigating injectivity phenomenon of $\pi_i(\alpha_{\mathrm{Loday}})$ for cylic group of order 2 (see Example 4.2 and Example 4.5).

We organise this paper as follow. We begin by review the pairing map defined by Loday in Section 2, which plays a key role in defining the assembly map. The main theorem is stated in Section 3, with examples about cyclic group of order 2 in Section 4.

1.1. **Notations.** Throughout this paper, we assume all rings are unital, associative, and satisfy the Invariant Basis Property. I.e., $R^n \cong R^m$ if and only if $n = m$. We always work with a discrete group G, and denote by BG its classifying space obtained from the bar construction.

By a spectrum, we mean a sequence $\mathbb{E} = \{\mathbb{E}_n\}_{n=0}^{\infty}$ of topological spaces, together with continuous maps

$$f_n : \mathbb{S}^1 \wedge \mathbb{E}_n \to \mathbb{E}_{n+1}.$$

For a ring R, we will use $\mathbb{K}_R^{\mathrm{GW}}$ to denote the Gersten-Wagoner non-connective algebraic K-theory spectrum (see [Ger72; Wag72] or Definition 2.12 below), whose n-th space is given by

$$\left(\mathbb{K}_R^{\mathrm{GW}}\right)_n := K_0(\Sigma^n R) \times BGL(\Sigma^n R)^+ ,$$

where $\Sigma^n R$ is the n-fold suspension of R, with the convention that $\Sigma^0 R := R$, and the superscript "+" denotes Quillen's Plus Construction. Its homotopy groups

$$K_i(R) := \pi_i\left(\mathbb{K}_R^{\mathrm{GW}}\right)$$

are the algebraic K-groups of the ring R. At last but not least, if $u \in GL(R)$, we denote by $\{u\} \in K_1(R)$ the element represented by u.

2. The Setups

2.1. **Loday's Pairing.** Let R, S be rings, and fix isomorphisms

$$\xi_{m,n} : R^m \otimes S^n \to (R \otimes S)^{mn} . \tag{2.1.1}$$

In this setting, the tensor product of matrices gives a group homomorphism

$$GL(m, R) \times GL(n, S) \to GL(mn, R \otimes S) ,$$

sending elementary matrices to elementary matrices. Hence we have an induced map

$$f_{m,n}^{R,S} : BGL(m, R)^+ \times BGL(n, S)^+ \to BGL(mn, R \otimes S)^+ . \tag{2.1.2}$$

Write

$$i_{mn} : BGL(mn, R \otimes S)^+ \to BGL(R \otimes S)^+$$

to be map induced by the canonical inclusion

$$GL(mn, R \otimes S) \to GL(R \otimes S) .$$

We then define

Definition 2.1 (The map $\gamma_{m,n}$, [Lod76, page 332])
The map

$$\gamma_{m,n} : BGL(m, R)^+ \times BGL(n, S)^+ \to BGL(R \otimes S)^+ \tag{2.1.3}$$

is defined by the formula:

$$\gamma_{m,n}(x,y) := i_{mn} \circ f^{R,S}_{m,n}(x,y) - i_{mn} \circ f^{R,S}_{m,n}(x,y_0) - i_{mn} \circ f^{R,S}_{m,n}(x_0,y)\,. \tag{2.1.4}$$

Here, the minus sign on the RHS comes from the H-group structure of $BGL(R\otimes S)^+$; and x_0 (resp. y_0) is the base-point in $BGL(m,R)^+$ (resp. $BGL(n,S)^+$). (I.e., represented by the $m\times m$ identity matrix.)

The choice of the isomorphisms $\xi_{m,n}$ says the map $\gamma_{m,n}$ is well-defined up to **weak-homotopy**. More precisely, the map $\gamma_{m,n}$ is well-defined only when restricted onto compact subsets of $BGL(m,R)^+ \times BGL(n,S)^+$.

One gets a map

$$\gamma : BGL(R)^+ \times BGL(S)^+ \to BGL(R\otimes S)^+ \tag{2.1.5}$$

by letting $m,n\to\infty$. This map, again, is well-defined up to weak-homotopy. Moreover, it is homotopy trivial on the wedge product $BGL(R)^+ \vee BGL(S)^+$.

Definition 2.2 (The Loday pairing map γ_{Loday}, [Lod76, Section 2.1.7 on page 333])
The map

$$\gamma_{\text{Loday}} : BGL(R)^+ \wedge BGL(S)^+ \to BGL(R\otimes S)^+ \tag{2.1.6}$$

is defined to be the filler of the following homotopy commutative diagram:

$$\begin{tikzcd} BGL(R)^+ \times BGL(S)^+ \arrow[r, "\gamma"] \arrow[d, "\text{proj}"'] & BGL(R\otimes S)^+ \\ BGL(R)^+ \wedge BGL(S)^+. \arrow[ur, "\gamma_{\text{Loday}}"'] & \end{tikzcd} \tag{2.1.7}$$

This allows us to define a multiplicative structure on algebraic K-theory.

Definition 2.3 (Loday product $\star$, [Lod76, Section 2.1.10 on page 335])
For each integers $i,j\geq 1$, we define the product

$$\star : K_i(R)\otimes K_j(S) \to K_{i+j}(R\otimes S) \tag{2.1.8}$$

by

$$[f]\star[g] := [\gamma_{\text{Loday}} \circ (f\wedge g)]\,, \tag{2.1.9}$$

where f and g are spheroids

$$f : \mathbb{S}^i \to BGL(R)^+\,,$$
$$g : \mathbb{S}^j \to BGL(S)^+\,.$$

Proposition 2.4 (Properties of the Loday Product $\star$)
The product map $\star$ is

(i) ([Lod76, Theorem 2.1.11 on page 335]) natural in R and S, associative and bilinear;
(ii) ([Lod76, Theorem 2.1.12 on page 335]) graded commutative:

$$[f] \star [g] = (-1)^{ij} ([g] \star [f]) \tag{2.1.10}$$

whenever $[f] \in K_i(R)$ and $[g] \in K_j(S)$;
(iii) ([Lod76, Proposition 2.2.3 on page 337]) the product $\{u\} \star \{v\}$ is the inverse of the Steinberg symbol:

$$\{u\} \star \{v\} = -\{u, v\}_{\mathrm{St}} \tag{2.1.11}$$

when $i = j = 1$. Here, we write the Abelian group $K_2(R \otimes S)$ additively.

2.2. **Extending the Pairing.** We want to modify the Loday pairing map

$$\gamma_{\mathrm{Loday}} : BGL(R)^+ \wedge BGL(S)^+ \to BGL(R \otimes S)^+$$

to a map

$$\gamma'_{\mathrm{Loday}} : K_R^+ \wedge K_S^+ \to K_{R\otimes S}^+,$$

where

$$K_R^+ := K_0(R) \times BGL(R)^+ . \tag{2.2.1}$$

This will yield an extended external product

$$K_i(R) \otimes K_j(S) \to K_{i+j}(R \otimes S)$$

for all i, $j \geq 0$. Moreover, this external product should be coherent with the isomorphism

$$K_i(R) \xrightarrow{\cong} K_{i+1}(\Sigma R) \tag{2.2.2}$$

promised by the **Gersten-Wagoner delooping** [Ger72; Wag72]:

$$K_R^+ \simeq \Omega K_{\Sigma R}^+. \tag{2.2.3}$$

At this point we need the closed form of the isomorphism in Equation (2.2.2). Define the element the element $\tau \in GL\left(\Sigma\mathbb{Z}\right) = GL\left(1, \Sigma\mathbb{Z}\right)$ by

$$\tau := \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \cdots \\ 1 & 0 & 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 1 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 1 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}. \tag{2.2.4}$$

The isomorphism in Equation (2.2.2) is given by

$$\begin{cases} [f] \in K_i(R) \mapsto [f] \star \{\tau\} & \text{if } i \geq 1 \text{ ([Lod76, Corollary 2.3.6])}, \\ [P] \in K_0(R) \mapsto [P]\#\{\tau\} := \{p \otimes \tau + (1-p) \otimes 1\} & \text{if } i = 0 \text{ ([Lod76, page 328])}, \end{cases} \tag{2.2.5}$$

where p is the projection operator associated to the finitely generated projective R-modules P. Equally important, the extended external product should be related to the classical products

$$\# : K_i(R) \otimes K_j(S) \to K_{i+j}(R \otimes S) \tag{2.2.6}$$

defined by Milnor [Mil72] for i, $j \geq 0$, and $i + j \leq 2$.

There is an obvious way to extend the pairing—tensor product of modules gives a group homomorphism

$$K_0(R) \otimes K_0(S) \to K_0(R \otimes S).$$

So one might use this to extend the pairing map

$$\gamma_{\text{Loday}} : BGL(R)^+ \wedge BGL(S)^+ \to BGL(R \otimes S)^+$$

by defining

$$\begin{aligned}\left(K_0(R) \times BGL(R)^+\right) \wedge \left(K_0(S) \times BGL(S)^+\right) &\to K_0(R \otimes S) \times BGL(R \otimes S)^+ \\ ([P], x) \wedge ([Q], y) &\mapsto ([P \otimes Q], \gamma_{\text{Loday}}(x, y)).\end{aligned}$$

This does not work well, because the induced product map

$$K_0(R) \otimes K_i(S) \to K_i(R \otimes S)$$

is the zero map for all $i > 0$.

Here, we propose a method to extend the original pairing and examine its properties.

Definition 2.5 (The modified Loday pairing γ'_{Loday})
The map

$$\gamma'_{\text{Loday}} : K_R^+ \wedge K_S^+ \to K_{R\otimes S}^+$$

is defined to be the filler of the following homotopy commutative diagram:

$$\begin{tikzcd} K_R^+ \wedge K_S^+ \arrow[r, dashed, "\gamma'_{\text{Loday}}"] \arrow[d, "\simeq"'] & K_{R\otimes S}^+ \arrow[d, "\simeq"] \\ \Omega BGL(\Sigma R)^+ \wedge \Omega BGL(\Sigma S)^+ \arrow[r, "\Omega^2 \gamma_{\text{Loday}}"'] & \Omega^2 BGL\left(\Sigma^2 (R \otimes S)\right)^+. \end{tikzcd} \tag{2.2.7}$$

Furthermore, we define a new product map

$$\star' : K_i(R) \otimes K_j(S) \to K_{i+j}(R \otimes S), \tag{2.2.8}$$

for all $i, j \geq 0$ by

$$[f] \star' [g] := \left[\gamma'_{\text{Loday}} \circ (f \wedge g)\right], \tag{2.2.9}$$

Corollary 2.6 (Cf [Lod76, Proposition 2.1.8 on page 334])
Let R, S be rings, the pairing map

$$\gamma'_{\mathrm{Loday}} : K_R^+ \wedge K_S^+ \to K_{R\otimes S}^+$$

defined in Definition 2.5 is

(i) natural in R and S;
(ii) bilinear;
(iii) associative;
(iv) commutative;

up to weak homotopy.

The two products are related in the following way:

Proposition 2.7
For each $i, j \geq 1$, if $[f] \in K_i(R)$ and $[g] \in K_j(S)$, then

$$[f] \star' [g] = [f] \star \left((-1)^j [g]\right) . \tag{2.2.10}$$

In particular, for $\{u\} \in K_1(R)$ and $\{v\} \in K_1(S)$, the $\star'$-product is given by the Steinberg symbol

$$\{u\} \star' \{v\} = \{u, v\}_{\mathrm{St}} . \tag{2.2.11}$$

Proof. We consider the following commutative diagram

$$\begin{tikzcd}
K_i(R) \otimes K_j(S) \arrow[rr, "\star'"] \arrow[d, "(-\star\{\tau\})\otimes(-\star\{\tau\})"', "\cong"] & & K_{i+j}(R\otimes S) \arrow[d, "\cong"', "(-\star\{\tau\})\star\{\tau\}"] \\
K_{i+1}(\Sigma R) \otimes K_{j+1}(\Sigma S) \arrow[rr, "\star"'] & & K_{i+j+2}\left(\Sigma^2\left(R\otimes S\right)\right)
\end{tikzcd} \tag{2.2.12}$$

induced by Diagram (2.2.7).

If $[f] \in K_i(R)$ and $[g] \in K_j(S)$, then the lower part of the diagram gives

$$\begin{aligned}
([f] \star \{\tau\}) \star ([g] \star \{\tau\}) &= [f] \star (\{\tau\} \star [g]) \star \{\tau\} && \text{(associativity)} \\
&= [f] \star \left((-1)^j \left([g] \star \{\tau\}\right)\right) \star \{\tau\} && \text{(graded-commutativity)} \\
&= \left([f] \star \left((-1)^j [g]\right) \star \{\tau\}\right) \star \{\tau\} . && \text{(associativity)}
\end{aligned}$$

So we must have

$$[f] \star' [g] = [f] \star \left((-1)^j [g]\right) .$$

□

2.3. **Properties of the Extended Pairing.**

2.3.1. *Relationship with the Classical Pairings in Algebraic K-theory.* Milnor defined multiplication structure on lower K-groups

$$\# : K_i(R) \otimes K_j(R) \to K_{i+j}(R \otimes R') \tag{2.3.1}$$

for i, $j \geq 0$, and $i + j \leq 2$ in [Mil72]. Actually, Milnor only defined the multiplication internally (i.e., when $R = R'$). But one can mimic his definition to obtain an external product as in Equation (2.3.1), so that when $R = R'$ it becomes Milnor's version.

The multiplication in Equation (2.3.1) is given by the formula:

$$\begin{cases} [P]\#[Q] &= [P \otimes Q] & \text{if } i = j = 0, \\ [P]\,\#\,\{u\} &= \{p \otimes u + (1-p) \otimes 1\} & \text{if } i = 0, \text{ and } j = 1, \\ \{u\}\,\#\,\{v\} &= \{u, v\}_{\mathrm{St}} & \text{if } i = j = 1. \end{cases} \tag{2.3.2}$$

We omit the case $i = 0$, $j = 2$ here. The main tool we need is

Proposition 2.8 ([Mil72, Lemma 8.9 on page 70])
The multiplication $\#$ in (2.3.1) is associative and bilinear for i, $j > 0$, and $i + j \leq 2$.

It is also straight forward to verify that $\#$ is graded-commutative. We now relate $\#$ and $\star'$.

Theorem 2.9
For every $[P] \in K_0(R)$, $[Q] \in K_0(R')$, $\{u\} \in K_1(R)$, and $\{v\} \in K_1(R')$, we have

(i) $[P] \star' [Q] = [P]\#[Q]$,

(ii) $[P] \star' \{v\} = [P]\#(-\{v\})$,

(iii) $\{u\} \star' \{v\} = \{u\}\,\#\,\{v\}$.

The author, however, does not know if the product

$$\star' : K_0(R) \otimes K_2(S) \to K_2(R \otimes S) \tag{2.3.3}$$

is compatible with Milnor's product

$$\# : K_0(R) \otimes K_2(S) \to K_2(R \otimes S). \tag{2.3.4}$$

defined in [Mil72, page 51].

Proof. We refer readers to Equation (2.2.5), and the commutative diagram in Equation (2.2.12).

(i) When $i = j = 0$, the diagram in (2.2.12) becomes

$$\begin{tikzcd} K_0(R) \otimes K_0(R') \arrow[r, "\star'"] \arrow[d, "(-\#\{\tau\}) \otimes (-\#\{\tau\})"', "\cong"] & K_0(R \otimes R') \arrow[d, "\cong"', "((-\#\{\tau\}) \star \{\tau\})"] \\ K_1(\Sigma R) \otimes K_1(\Sigma R') \arrow[r, "\star"'] & K_2(\Sigma^2(R \otimes R')). \end{tikzcd} \tag{2.3.5}$$

We then compute

$$
\begin{aligned}
([P]\#\{\tau\})\star([Q]\#\{\tau\}) &= -\{[P]\#\{\tau\},[Q]\#\{\tau\}\}_{\mathrm{St}} && \text{(Equation (2.1.11))}\\
&= \{[P]\#\{\tau\},(-[Q])\#\{\tau\}\}_{\mathrm{St}} && \text{(Bilinearity of } \{-,-\}_{\mathrm{St}})\\
&= ([P]\#\{\tau\})\#((-[Q])\#\{\tau\}) && \text{(Equation (2.3.2))}\\
&= [P]\#(\{\tau\}\#(-[Q]))\#\{\tau\} && \text{(Associativity of \#)}\\
&= [P]\#(-[Q]\#\{\tau\})\#\{\tau\} && \text{(Graded-commutativity of \#)}\\
&= -([P]\#([Q]\#\{\tau\})\#\{\tau\}) && \text{(Bilinearity of \#)}\\
&= -\{[P]\#([Q]\#\{\tau\}),\tau\}_{\mathrm{St}} && \text{(Equation (2.3.2))}\\
&= ([P]\#([Q]\#\{\tau\}))\star\{\tau\} && \text{(Equation (2.1.11))}\\
&= (([P]\#[Q])\#\{\tau\})\star\{\tau\}, && \text{(Associativity of \#)}
\end{aligned}
$$

proving

$$[P]\star'[Q] = [P]\#[Q].$$

(ii) When $i=0$, $j=1$, Diagram (2.2.12) becomes

$$
\begin{tikzcd}
K_0(R)\otimes K_1(R') \arrow[r, "\star'"] \arrow[d, "(-\#\{\tau\})\otimes(-\star\{\tau\})"', "\cong"] & K_1(R\otimes R') \arrow[d, "((-\star\{\tau\})\star\{\tau\})", "\cong"'] \\
K_1(\Sigma R)\otimes K_2(\Sigma R') \arrow[r, "\star"'] & K_3(\Sigma^2(R\otimes R')).
\end{tikzcd}
\tag{2.3.6}
$$

We then compute

$$
\begin{aligned}
([P]\#\{\tau\})\star(\{v\}\star\{\tau\}) &= (([P]\#\{\tau\})\star\{v\})\star\{\tau\} && \text{(Associativity of } \star)\\
&= (-\{[P]\#\{\tau\},v\}_{\mathrm{St}})\star\{\tau\} && \text{(Proposition 2.4)}\\
&= (-(([P]\#\{\tau\})\#\{v\}))\star\{\tau\} && \text{(Equation (2.3.2))}\\
&= (-([P]\#(\{\tau\}\#\{v\})))\star\{\tau\} && \text{(Associativity of \#)}
\end{aligned}
$$

$$
\begin{aligned}
&= ([P]\#(\{v\}\#\{\tau\}))\star\tau && \text{(Graded-commutativity of \#)}\\
&= (([P]\#\{v\})\#\{\tau\})\star\{\tau\} && \text{(Associativity of \#)}\\
&= \{[P]\#\{v\},\{\tau\}\}_{\mathrm{St}}\star\{\tau\} && \text{(Equation (2.3.2))}\\
&= (-(([P]\#\{v\})\star\{\tau\}))\star\{\tau\} && \text{(Proposition 2.4)}\\
&= ((-([P]\#\{v\})\star\{\tau\}))\star\{\tau\} && \text{(Bilinearity of } \star)\\
&= (([P]\#(-\{v\}))\star\{\tau\})\star\{\tau\} && \text{(Bilinearity of \#)},
\end{aligned}
$$

proving

$$[P]\star'\{v\} = [P]\#(-\{v\}).$$

(iii)

$$
\begin{aligned}
\{u\}\star'\{v\} &= \{u,v\}_{\mathrm{St}} && \text{(Proposition 2.7)}\\
&= \{u\}\#\{v\}.
\end{aligned}
$$

□

2.3.2. *The Structure Maps of the Non-connective Gersten-Wagoner Algebraic K-theory Spectrum.* We relate our modified pairing

$$\gamma'_{\mathrm{Loday}}: K_R^+\wedge K_S^+\to K_{R\otimes S}^+$$

to the structure map

$$\mathbb{S}^1\wedge K_R^+\to K_{\Sigma R}^+$$

of the Gersten-Wagoner spectrum $\mathbb{K}_R^{\mathrm{GW}}$. This will be used later in proving our version of the Loday assembly is a map of spectra.

In [Lod76, page 341–343], Loday gave an explicit description of the Gersten-Wagoner delooping

$$K_R^+\xrightarrow{\simeq}\Omega K_{\Sigma R}^+,$$

so that the induced isomorphisms on homotopy groups in Equation (2.2.2) are given as in Equation (2.2.5). We will use our modified pairing to define a map

$$\mathbb{S}^1\wedge K_R^+\to K_{\Sigma R}^+,$$

whose adjoint

$$K_R^+ \to \Omega K_{\Sigma R}^+$$

will induce isomorphisms on homotopy groups, and hence is a homotopy equivalence by the Whitehead Theorem. We begin by thinking the circle $\mathbb{S}^1$ as the classifying space $B\langle t\rangle$ of the infinite cyclic group generated by t, and define the following group homomorphism:

Definition 2.10 (The map $\mathbf{t}^+ : B\langle t\rangle \to K_{\Sigma\mathbb{Z}}^+$)
Let $\langle t\rangle$ be the infinite cyclic group generated by t, and $\tau \in GL(\Sigma\mathbb{Z})$ be the element defined in Equation (2.2.4). Define the group homomorphism

$$\begin{aligned} \langle t\rangle &\to GL(\Sigma\mathbb{Z}) \\ t &\mapsto \tau^{-1}. \end{aligned}$$

The induced map

$$B\langle t\rangle \to K_{\Sigma\mathbb{Z}}^+ \tag{2.3.7}$$

is denoted by $\mathbf{t}^+$.

Proposition 2.11 (Another Gersten-Wagoner Delooping)
Let R be a ring, and $\langle t\rangle$ be the infinite cyclic group generated by t. The adjoint of the composition

$$B\langle t\rangle \wedge K_R^+ \xrightarrow{\mathbf{t}^+\wedge \mathrm{id}} K_{\Sigma\mathbb{Z}}^+ \wedge K_R^+ \xrightarrow{\gamma'_{\mathrm{Loday}}} K_{\Sigma R}^+ \tag{2.3.8}$$

is a weak equivalence, and hence a homotopy equivalence.

Proof. A class $[f] \in K_i(R)$ is represented by a spheroid

$$f : \mathbb{S}^i \to K_R^+.$$

Suspending it yields the following composition

$$\mathbb{S}^1 \wedge \mathbb{S}^i \xrightarrow{\mathrm{id}\wedge f} \mathbb{S}^1 \wedge K_R^+ \xrightarrow{\mathbf{t}^+\wedge \mathrm{id}} K_{\Sigma\mathbb{Z}}^+ \wedge K_R^+ \xrightarrow{\gamma'_{\mathrm{Loday}}} K_{\Sigma R}^+, \tag{2.3.9}$$

which represents the element $[f] \star' \left\{\tau^{-1}\right\} \in K_{i+1}(\Sigma R)$. In particular, we have

$$[f] \star' \left\{\tau^{-1}\right\} = \begin{cases} [f] \star \{\tau\} & \text{if } i > 0 \quad \text{(Proposition 2.7)} \\ [f] \# \{\tau\} & \text{if } i = 0 \quad \text{(Theorem 2.9 (ii))} \end{cases} \tag{2.3.10}$$

The map on homotopy groups

$$K_i(R) \to K_{i+1}(\Sigma R)$$

induced by the adjoint of (2.3.8) sends

$$[f] \mapsto [f] \star' \left\{\tau^{-1}\right\},$$

which is an isomorphism for $i = 0$ (resp. $i > 0$) by [Lod76, after Theorem 1.4.7 on page 328] (resp. [Lod76, Corollary 2.3.6 on page 345]). Therefore, Whitehead Theorem asserts the homotopy equivalence

$$K_R^+ \simeq K_{\Sigma R}^+.$$

□

Consequently, we have the following Ω-spectrum

Definition 2.12 (The Non-connective Gersten-Wagoner Algebraic K-theory Spectrum $\mathbb{K}_R^{\mathrm{GW}}$)
Let R be a ring. Define the spectrum $\mathbb{K}_R^{\mathrm{GW}}$ by having n-th space as

$$\left(\mathbb{K}_R^{\mathrm{GW}}\right)_n := \begin{cases} K_{\Sigma^n R}^+ & \text{if } n \geq 0, \\ \Omega^{-n} K_R^+ & \text{if } n < 0, \end{cases} \tag{2.3.11}$$

with the convention that $\Sigma^0 R := R$. The structure maps

$$f_n : \left(\mathbb{K}_R^{\mathrm{GW}}\right)_n \to \left(\mathbb{K}_R^{\mathrm{GW}}\right)_{n+1} \tag{2.3.12}$$

are given by

$$f_n := \begin{cases} (2.3.8) & \text{if } n \geq 0, \\ \text{adjoint of } \mathrm{id}_{\left(\mathbb{K}_R^{\mathrm{GW}}\right)_n} & \text{if } n < 0. \end{cases} \tag{2.3.13}$$

By construction, $\mathbb{K}_R^{\mathrm{GW}}$ is an Ω-spectrum.

3. The Main Theorem

3.1. **Definition of the Loday Assembly.** We use the modified Loday pairing γ'_{Loday} to define the Loday assembly map:

Definition 3.1 (The Loday Assembly Map α_{Loday} in Reduced Case)
Let R be a ring, and G be a group. The **Loday assembly map** is the map of spectra

$$\alpha_{\mathrm{Loday}} : BG \wedge \mathbb{K}_R^{\mathrm{GW}} \to \mathbb{K}_{R[G]}^{\mathrm{GW}}, \tag{3.1.1}$$

whose components are given by the composition of maps of spaces:

$$BG \wedge K_{\Sigma^m R}^+ \xrightarrow{j^+ \wedge \mathrm{id}} K_{\mathbb{Z}[G]}^+ \wedge K_{\Sigma^m R}^+ \xrightarrow{\gamma'_{\mathrm{Loday}}} K_{\Sigma^m R[G]}^+. \tag{3.1.2}$$

Here, the map

$$j^+ : BG \to K_{\mathbb{Z}[G]}^+ \tag{3.1.3}$$

is induced by the group homomorphism

$$\begin{aligned} j : G &\to GL\left(\mathbb{Z}[G]\right) \\ g &\mapsto \begin{bmatrix} g & 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & 0 & \cdots \\ 0 & 0 & 1 & 0 & \cdots \\ 0 & 0 & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}. \end{aligned} \tag{3.1.4}$$

We need to justify our definition.

Proposition 3.2 (Cf [Lod76, Proposition 4.1.1 on page 356])
Let R be a ring, and G be a group. The Loday assembly map

$$\alpha_{\text{Loday}} : BG \wedge \mathbb{K}_R^{\text{GW}} \to \mathbb{K}_{R[G]}^{\text{GW}}$$

defined in Definition 3.1 is a map of spectra.

Proof. We need to show the components of α_{Loday}, as defined in Equation (3.1.2), commute with the structure maps of $\mathbb{K}_R^{\text{GW}}$ as defined in Equation (2.3.8). Let $\langle t \rangle$ be the infinite cyclic group generated by t. By unwrapping the definitions of α_{Loday} and the structure maps given in Proposition 2.11, the following diagram

$$\begin{tikzcd}
BG \wedge B\langle t\rangle \wedge K_R^+ \arrow[r, "j^+ \wedge \mathrm{id} \wedge \mathrm{id}"] \arrow[d, "\mathrm{id} \wedge \mathrm{t}^+ \wedge \mathrm{id}"'] & K_{\mathbb{Z}[G]}^+ \wedge B\langle t\rangle \wedge K_R^+ \arrow[r, "\text{twist}", "\cong"'] \arrow[d, "\mathrm{id} \wedge \mathrm{t}^+ \wedge \mathrm{id}"'] & B\langle t\rangle \wedge K_{\mathbb{Z}[G]}^+ \wedge K_R^+ \arrow[r, "\mathrm{id} \wedge \gamma'_{\text{Loday}}"] \arrow[d, "\mathrm{t}^+ \wedge \mathrm{id} \wedge \mathrm{id}"'] & B\langle t\rangle \wedge K_{R[G]}^+ \arrow[d, "\mathrm{t}^+ \wedge \mathrm{id}"] \\
BG \wedge K_{\Sigma\mathbb{Z}}^+ \wedge K_R^+ \arrow[r, "j^+ \wedge \mathrm{id} \wedge \mathrm{id}"] \arrow[d, "\mathrm{id} \wedge \gamma'_{\text{Loday}}"'] & K_{\mathbb{Z}[G]}^+ \wedge K_{\Sigma\mathbb{Z}}^+ \wedge K_R^+ \arrow[r, "\cong", "\text{twist}"'] \arrow[d, "\mathrm{id} \wedge \gamma'_{\text{Loday}}"'] & K_{\Sigma\mathbb{Z}}^+ \wedge K_{\mathbb{Z}[G]}^+ \wedge K_R^+ \arrow[r, "\mathrm{id} \wedge \gamma'_{\text{Loday}}"] \arrow[d, "\gamma'_{\text{Loday}} \circ \gamma'_{\text{Loday}}"'] & K_{\Sigma\mathbb{Z}}^+ \wedge K_{R[G]}^+ \arrow[dl, "\gamma'_{\text{Loday}}"] \\
BG \wedge K_{\Sigma R}^+ \arrow[r, "j^+ \wedge \mathrm{id} \wedge \mathrm{id}"'] & K_{\mathbb{Z}[G]}^+ \wedge K_{\Sigma R}^+ \arrow[r, "\gamma'_{\text{Loday}}"'] & K_{\Sigma R[G]}^+ &
\end{tikzcd} \tag{3.1.5}$$

commutes up to weak homotopy by Corollary 2.6. Here, the columns are the structure maps, and the rows are components of the Loday assembly. ☐

The homotopy groups $\pi_i(BG \wedge \mathbb{K}_R^{\text{GW}})$ of the spectrum $BG \wedge \mathbb{K}_R^{\text{GW}}$ is the reduced homology groups of the classifying space BG of G with coefficients in $\mathbb{K}_R^{\text{GW}}$. So the induced map

$$\pi_i(\alpha_{\text{Loday}}) : \pi_i(BG \wedge \mathbb{K}_R^{\text{GW}}) \to K_i(R[G])$$

approximates the algebraic K-theory of the group ring $R[G]$ by a reduced homology theory. We want to lift the domain to an unreduced homology theory. Recall for any (based) topological space X, there is a cofibre sequence

$$X \to X_+ \to \mathbb{S}^0 \tag{3.1.6}$$

of spaces, where $X_+ := X \sqcup \text{point}$ is the space obtained by adding a disjoint point to X. If $\mathbb{E}$ is a spectrum, then we have a cofibre sequence

$$X \wedge \mathbb{E} \to X_+ \wedge \mathbb{E} \to \mathbb{S}^0 \wedge \mathbb{E} \tag{3.1.7}$$

of spectra. Consequently, we have a splitting

$$\begin{aligned} X_+ \wedge \mathbb{E} &\simeq (X \wedge \mathbb{E}) \vee \left(\mathbb{S}^0 \wedge \mathbb{E}\right) \\ &\cong (X \wedge \mathbb{E}) \vee \mathbb{E} \end{aligned} \tag{3.1.8}$$

of spectra. This is how we extend the Loday assembly to the unreduced case.

Definition 3.3 (The Loday Assembly Map α_{Loday} in Unreduced Case) Let R be a ring and G be a group. The canonical inclusion $R \to R[G]$ induces a map

$$i : \mathbb{K}_R^{\text{GW}} \to \mathbb{K}_{R[G]}^{\text{GW}} \tag{3.1.9}$$

of spectra. The Loday assembly in unreduced case is defined by

$$BG_+ \wedge \mathbb{K}_R^{\text{GW}} \simeq \left(BG \wedge \mathbb{K}_R^{\text{GW}}\right) \vee \mathbb{K}_R^{\text{GW}} \xrightarrow{\text{Definiton } 3.1 \vee i} \mathbb{K}_{R[G]}^{\text{GW}}. \tag{3.1.10}$$

By abuse of terminology, we call this map the Loday assembly and denote it by α_{Loday} as well

3.2. **A Subgroup of the Source of Loday Assembly.** The source of the Loday assembly

$$\pi_i(\alpha_{\text{Loday}}) : \pi_{i+1}\left(BG_+ \wedge \mathbb{K}_R^{\text{GW}}\right) \to K_{i+1}\left(R[G]\right) \tag{3.2.1}$$

on homotopy groups can be computed by the Atiyah-Hirzebruch spectral sequence (AHSS):

$$\begin{aligned} \text{AHSS}\left(BG\right)^2_{p,q} &\cong H_p\left(BG; K_q(R)\right) \\ &\Rightarrow \pi_{p+q}\left(BG_+ \wedge \mathbb{K}_R^{\text{GW}}\right), \end{aligned} \tag{3.2.2}$$

and in particular,

$$\begin{aligned} E^2_{1,i} &\cong H_1\left(BG; K_i(R)\right) \\ &\cong G_{ab} \otimes K_i(R). \end{aligned} \tag{3.2.3}$$

Because the spectral sequence (3.2.3) is concetrated on the right-half plane, it follows that the differential

$$d^2_{1,i} : E^2_{1,i} \to E^2_{-1,i+1} \tag{3.2.4}$$

has to be zero. So the group $E^\infty_{1,i}$ is a quotient of $E^2_{1,i}$, and we shall see it is also a subgroup of the source $\pi_{i+1}\left(BG_+ \wedge \mathbb{K}_R^{\text{GW}}\right)$.

Fix a spectrum $\mathbb{E}$. For any CW complex X, we have a homeomorphism

$$X_+ \cong X \vee \mathbb{S}^0, \tag{3.2.5}$$

and hence an isomorphism of homotopy groups

$$\pi_{i+1}(X_+ \wedge \mathbb{E}) \cong \pi_{i+1}(X \wedge \mathbb{E}) \oplus \pi_{i+1}(\mathbb{S}^0 \wedge \mathbb{E}). \tag{3.2.6}$$

Fix a skeleta filtration

$$\text{point} = X^{(0)} \subseteq X^{(1)} \subseteq \cdots \subseteq X \tag{3.2.7}$$

on X, the identification in Equation (3.2.5) gives a skeleta filtration

$$X_+^{(p)} \cong X^{(p)} \vee \mathbb{S}^0 \tag{3.2.8}$$

on X_+. Thus we have a splitting for the AHSS

$$\text{AHSS}\left(X_+\right)^r_{p,q} \cong \text{AHSS}\left(X\right)^r_{p,q} \oplus \text{AHSS}\left(\mathbb{S}^0\right)^r_{p,q} \tag{3.2.9}$$

for $r \geq 1$. Tracing through the construction of the AHSS (see, for example [Wic15]), we see that the splitting in Equation 3.2.5 respects the filtration on homotopy groups:

$$\mathcal{F}_1\left[\pi_{i+1}(X_+ \wedge \mathbb{E})\right] \cong \mathcal{F}_1\left[\pi_{i+1}(X \wedge \mathbb{E})\right] \oplus \mathcal{F}_1\left[\pi_{i+1}(\mathbb{S}^0 \wedge \mathbb{E})\right], \tag{3.2.10}$$

for which

$$\mathcal{F}_1\left[\pi_{i+1}\left(X_+ \wedge \mathbb{E}\right)\right] := \operatorname{im}\left(\pi_{i+1}\left(X_+^{(1)} \wedge \mathbb{E}\right) \to \pi_{i+1}\left(X_+ \wedge \mathbb{E}\right)\right). \tag{3.2.11}$$

The subgroup

$$\mathcal{F}_1\left[\pi_{i+1}(X_+ \wedge \mathbb{E})\right] \subseteq \pi_{i+1}(X_+ \wedge \mathbb{E}) \tag{3.2.12}$$

is what we are interested in, and it can be identified as:

$$\begin{aligned}\mathcal{F}_1\left[\pi_{i+1}(X_+ \wedge \mathbb{E})\right] &\cong \mathcal{F}_1\left[\pi_{i+1}(X \wedge \mathbb{E})\right] \oplus \mathcal{F}_1\left[\pi_{i+1}(\mathbb{S}^0 \wedge \mathbb{E})\right] \\ &= \mathrm{AHSS}\,(X)^{\infty}_{1,i} \oplus \pi_{i+1}(\mathbb{E}).\end{aligned} \tag{3.2.13}$$

3.3. **Statement of the Main Theorem and the Proof.** Now, let us work with $\mathbb{E} = \mathbb{K}_R^{\mathrm{GW}}$, and $X = BG$. Then Equation (3.2.13) says

$$\begin{aligned}\mathcal{F}_1\left[\pi_{i+1}(BG_+ \wedge \mathbb{K}_R^{\mathrm{GW}})\right] &\cong \mathcal{F}_1\left[\pi_{i+1}(BG \wedge \mathbb{K}_R^{\mathrm{GW}})\right] \oplus \mathcal{F}_1\left[\pi_{i+1}(\mathbb{S}^0 \wedge \mathbb{K}_R^{\mathrm{GW}})\right] \\ &= \mathrm{AHSS}\,(BG)^{\infty}_{1,i} \oplus \pi_{i+1}(\mathbb{K}_R^{\mathrm{GW}}) \\ &= \mathrm{AHSS}\,(BG)^{\infty}_{1,i} \oplus K_{i+1}(R).\end{aligned} \tag{3.3.1}$$

So the source of the Loday assembly

$$\alpha_{\mathrm{Loday}} : \pi_{i+1}(BG_+ \wedge \mathbb{K}_R^{\mathrm{GW}}) \to K_{i+1}(R[G])$$

contains

$$\mathrm{AHSS}\,(BG)^{\infty}_{1,i} \oplus K_{i+1}(R)$$

as a subgroup. In fact, we already know what the Loday assembly is doing when restricted onto the summand $K_{i+1}(R)$—it is induced by the inclusion

$$R \to R[G]$$

of rings. See [Lod76, before Lemma 4.1.2 on page 356], or Definition 3.3 before. What we want now is to study the Loday assembly when restricted onto the summand $\mathrm{AHSS}\,(BG)^{\infty}_{1,i}$.

The splitting of the AHSS in Equation (3.2.9) says the differentials of $\mathrm{AHSS}\,(BG_+)^2_{p,q}$ splits into two parts, which we illustrate here:

$$
\underbrace{\begin{array}{c|cccc}
q & & & & \\
 & K_5(R) & H_1(BG;K_5(R)) & H_2(BG;K_5(R)) & \cdots \\
 & K_4(R) & H_1(BG;K_4(R)) & H_2(BG;K_4(R)) & \cdots \\
 & K_3(R) & H_1(BG;K_3(R)) & H_2(BG;K_3(R)) & \cdots \\
 & K_2(R) & H_1(BG;K_2(R)) & H_2(BG;K_2(R)) & \cdots \\
 & K_1(R) & H_1(BG;K_1(R)) & H_2(BG;K_1(R)) & \cdots \\
 & K_0(R) & H_1(BG;K_0(R)) & H_2(BG;K_0(R)) & \cdots \\
\hline
 & & & & p
\end{array}}_{\mathrm{AHSS}(BG_+)^2_{p,q}}
\cong
\underbrace{\begin{array}{c|cccc}
q & & & & \\
 & 0 & H_1(BG;K_5(R)) & H_2(BG;K_5(R)) & \cdots \\
 & 0 & H_1(BG;K_4(R)) & H_2(BG;K_4(R)) & \cdots \\
 & 0 & H_1(BG;K_3(R)) & H_2(BG;K_3(R)) & \cdots \\
 & 0 & H_1(BG;K_2(R)) & H_2(BG;K_2(R)) & \cdots \\
 & 0 & H_1(BG;K_1(R)) & H_2(BG;K_1(R)) & \cdots \\
 & 0 & H_1(BG;K_0(R)) & H_2(BG;K_0(R)) & \cdots \\
\hline
 & & & & p
\end{array}}_{\mathrm{AHSS}(BG)^2_{p,q}}
\oplus
\underbrace{\begin{array}{c|cccc}
q & & & & \\
 & K_5(R) & 0 & 0 & \cdots \\
 & K_4(R) & 0 & 0 & \cdots \\
 & K_3(R) & 0 & 0 & \cdots \\
 & K_2(R) & 0 & 0 & \cdots \\
 & K_1(R) & 0 & 0 & \cdots \\
 & K_0(R) & 0 & 0 & \cdots \\
\hline
 & & & & p
\end{array}}_{\mathrm{AHSS}(\mathbb{S}^0)^2_{p,q}}
\tag{3.3.2}
$$

(We omit the negative K-groups because they are annoying to include.) We remark that the groups $\mathrm{AHSS}\,(BG)^2_{0,q}$ trivial because the reduced homology $\widetilde{H}_0(BG)$ is trivial, for BG being path-connected. Consequently,

Lemma 3.4
Let G be a group, and R be a ring. The differentials

$$d^2_{1,q} : \mathrm{AHSS}\,(BG_+)^2_{1,q} \to \mathrm{AHSS}\,(BG_+)^2_{-1,q+1} \tag{3.3.3}$$

in the E^2-page of the AHSS of BG_+ associated to the K-theory spectrum $\mathbb{K}^{\mathrm{GW}}_R$ are trivial for all q.

Proof. The splitting of the AHSS in Equation (3.2.9) says the differential

$$d^2_{1,q} : \mathrm{AHSS}\,(BG_+)^2_{1,q} \to \mathrm{AHSS}\,(BG_+)^2_{-1,q+1}$$

splits into direct sum of the two differentials

$$\mathrm{AHSS}\,(BG)^2_{1,q} \to \mathrm{AHSS}\,(BG)^2_{-1,q+1}\,, \tag{3.3.4}$$

$$\mathrm{AHSS}\,(\mathbb{S}^0)^2_{1,q} \to \mathrm{AHSS}\,(\mathbb{S}^0)^2_{-1,q+1}\,. \tag{3.3.5}$$

From Diagram (3.3.2), we see that the first (resp. second) map is trivial because the codomain (resp. domain) is trivial, hence proving the claim. □

Lemma 3.4 that says the group $\mathrm{AHSS}\,(BG_+)^{\infty}_{1,i}$ is a quotient of $\mathrm{AHSS}\,(BG_+)^2_{1,i}$, and therefore we have a diagram

$$\begin{tikzcd}
\mathrm{AHSS}\,(BG)^2_{1,i} \cong G_{ab} \otimes K_i(R) \arrow[r, dashed] \arrow[d, two heads] & K_{i+1}\,(R[G]) \\
\mathrm{AHSS}\,(BG)^\infty_{1,i} \arrow[r, hook] & \pi_{i+1}\left(BG \wedge \mathbb{K}_R^{\mathrm{GW}}\right). \arrow[u, "\alpha_{\mathrm{Loday}}"']
\end{tikzcd} \tag{3.3.6}$$

We would like to study the filler. (Perhaps we should explain the absent of the disjoint base-point "+" in Diagram (3.3.6) is not a typo. As pointed out in Equation (3.3.1), $\pi_{i+1}(BG_+ \wedge \mathbb{K}_R^{\mathrm{GW}})$ contains $K_{i+1}(R) \oplus \mathrm{AHSS}\,(BG)^\infty_{1,i}$ as a subgroup, and we want to know what the Loday assembly is doing on the term $\mathrm{AHSS}\,(BG)^\infty_{1,i}$.)

Let us go back to the E^1-page of the AHSS. Unlike the later pages, the E^1-page is not homotopy invariant—it depends on the choice of cell structure on BG. **For our purpose, we are using the skeletal filtration on the bar construction for BG.** Then we have:

$$\mathrm{AHSS}\,(BG)^1_{1,i} \cong \pi_{i+1}\left(\left(\bigvee_{g\in G} \mathbb{S}^1\right) \wedge \mathbb{K}_R^{\mathrm{GW}}\right). \tag{3.3.7}$$

The filler we want to study is then induced by the composition

$$\pi_{i+1}\left(\left(\bigvee_{g\in G} \mathbb{S}^1\right) \wedge \mathbb{K}_R^{\mathrm{GW}}\right) \xrightarrow{(\mathfrak{i}_G)_*} \pi_{i+1}\left(BG \wedge \mathbb{K}_R^{\mathrm{GW}}\right) \xrightarrow{\alpha_{\mathrm{Loday}}} K_{i+1}\,(R[G]), \tag{3.3.8}$$

for which the first arrow is induced by the inclusion

$$\mathfrak{i}_G : \bigvee_{g\in G} \mathbb{S}^1 \hookrightarrow BG \tag{3.3.9}$$

of the 1-skeleton of BG into the BG. This follows from the construction of the AHSS. (See [Wic15] for example.) When $G = \langle t \rangle$ is the infinite cyclic group, we know the circle $\mathbb{S}^1$ is a model for $B\langle t \rangle$. Moreover,

Lemma 3.5
The inclusion

$$\mathfrak{i}_t : \mathbb{S}^1 \to B\langle t \rangle \tag{3.3.10}$$

that sends $\mathbb{S}^1$ to the 1-cell of $B\langle t \rangle$ labelled by the generator t is a weak equivalence.

Proof. It is clear that the induced group homomorphism

$$(\mathfrak{i}_t)_* : \pi_i\left(\mathbb{S}^1\right) \to \pi_i\left(B\langle t \rangle\right) \tag{3.3.11}$$

is an isomorphism for $i \neq 1$, because the source and target are both trivial. So we need to check the map

$$(\mathfrak{i}_t)_* : \pi_1\left(\mathbb{S}^1\right) \to \pi_1\left(B\langle t \rangle\right) \tag{3.3.12}$$

is an isomorphism.

The map in Equation (3.3.12) is just an endomorphism of infinite cyclic group. In particular, it is surjective by the very definition of $\mathfrak{i}_t$. (The generator is in the image.) So the First Isomorphism Theorem says the map in Equation (3.3.12) must have trivial kernel, and hence an isomorphism. ☐

The Whitehead Theorem then asserts $\mathfrak{i}_t$ is a homotopy equivalence. Thus when $G = \langle t \rangle$ is the infinite cyclic group, we have the following commutative diagram

$$\begin{tikzcd}
\bigvee\limits_{g\in\langle t\rangle} \mathbb{S}^1 \arrow[rr, "\mathfrak{i}_{\langle t\rangle}"] & & B\langle t\rangle \\
& & \mathbb{S}^1 \arrow[u, "\mathfrak{i}_t"', "\simeq"] \arrow[ull, "\mathfrak{i}_t"]
\end{tikzcd} \tag{3.3.13}$$

The upshot is that we are able to describe the composition in Equation (3.3.8) when $G = \langle t \rangle$.

Proposition 3.6
When $G = \langle t \rangle$ is the infinite cyclic group, the composition in Equation (3.3.8) can be identified as

$$\begin{tikzcd}
\pi_{i+1}\left(\left(\bigvee\limits_{g\in\langle t\rangle} \mathbb{S}^1\right) \wedge \mathbb{K}_R^{\mathrm{GW}}\right) \arrow[r, "(\mathfrak{i}_{\langle t\rangle})"] & \pi_{i+1}\left(B\langle t\rangle \wedge \mathbb{K}_R^{\mathrm{GW}}\right) \arrow[r, "\alpha_{\mathrm{Loday}}"] & K_{i+1}\left(R[t^{\pm 1}]\right) \\
\pi_{i+1}\left(\mathbb{S}^1 \wedge \mathbb{K}_R^{\mathrm{GW}}\right) \arrow[u, "(\mathfrak{i}_t)_*"] \arrow[r, "\mathrm{id}"'] & \pi_{i+1}\left(\mathbb{S}^1 \wedge \mathbb{K}_R^{\mathrm{GW}}\right) \arrow[u, "(\mathfrak{i}_t)_*", "\cong"'] \arrow[r] & K_{i+1}\left(R[t^{\pm 1}]\right) \arrow[u, equal] \\
K_i(R) \arrow[u, "\cong"] \arrow[r, "\mathrm{id}"'] & K_i(R) \arrow[u, "\cong"] \arrow[r] & K_{i+1}\left(R[t^{\pm 1}]\right), \arrow[u, equal]
\end{tikzcd} \tag{3.3.14}$$

where the bottom composition sends $[f] \in K_i(R)$ to

$$[f] \star \{\mu_{t^{-1}}\} \in K_{i+1}\left(R[t^{\pm 1}]\right),$$

for which $\mu_{t^{-1}} \in \mathrm{Aut}\left(R[t^{\pm 1}]\right)$ is multiplication by t^{-1}.

We remind readers the commutativity of the top-left square comes from Diagram (3.3.13).

Proof. The class $[f] \in K_i(R)$ is represented by the spheroid

$$f : \mathbb{S}^i \to K_R^+,$$

where

$$K_R^+ := K_0(R) \times BGL(R)^+ .$$

Taking suspension yields

$$\mathbb{S}^1 \wedge \mathbb{S}^i \xrightarrow{\mathrm{id} \wedge f} \mathbb{S}^1 \wedge K_R^+ \xrightarrow{j^+ \wedge \mathrm{id}} K_{\mathbb{Z}[t^{\pm 1}]}^+ \wedge K_R^+ \xrightarrow{\gamma'_{\mathrm{Loday}}} K_{R[t^{\pm 1}]}^+.$$

This composition represents the element $\alpha_{\mathrm{Loday}}([f]) \in K_{i+1}\left(R[t^{\pm 1}]\right)$, as well as the product

$$[f] \star' \left\{j^+\right\} \in K_{i+1}\left(R[t^{\pm 1}]\right).$$

Now, j^+ is induced by the group homomorphism

$$\begin{aligned} j : \langle t \rangle &\to GL\left(\mathbb{Z}[t^{\pm 1}]\right) \\ t &\mapsto \begin{bmatrix} t & 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & 0 & \cdots \\ 0 & 0 & 1 & 0 & \cdots \\ 0 & 0 & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}. \end{aligned}$$

So $\left\{j^+\right\} = \{\mu_t\}$, and thus

$$\begin{aligned} [f] \star' \left\{j^+\right\} &= [f] \star' \{\mu_t\} \\ &= [f] \star (-\{\mu_t\}) && \text{(Proposition 2.7)} \\ &= [f] \star \{\mu_{t^{-1}}\} \end{aligned}$$

as desired. □

For arbitrary group G, we recall the isomorphism

$$\begin{aligned} G &\cong \mathrm{Hom}(\langle t \rangle, G) \\ g &\mapsto (\varphi_g : t \mapsto g), \end{aligned} \tag{3.3.15}$$

allowing us to extend Proposition 3.6 to arbitrary G.

Proposition 3.7
For an arbitrary group G, under the canonical isomorphism

$$\pi_{i+1}\left(\left(\bigvee_{g \in G} \mathbb{S}^1\right) \wedge \mathbb{K}_R^{\mathrm{GW}}\right) \cong \bigoplus_{g \in G} K_i(R), \tag{3.3.16}$$

the composition in (3.3.8) can be identified as

$$\begin{tikzcd}
\pi_{i+1}\left(\left(\bigvee_{g\in G}\mathbb{S}^1\right)\wedge\mathbb{K}_R^{\mathrm{GW}}\right) \arrow[r, "(\mathfrak{i}_G)_*"] \arrow[d, phantom, "\wr\|"] & \pi_{i+1}\left(BG\wedge\mathbb{K}_R^{\mathrm{GW}}\right) \arrow[r, "\alpha_{\mathrm{Loday}}"] \arrow[d, phantom, "\|"] & K_{i+1}\left(R[G]\right) \arrow[d, phantom, "\|"] \\
\bigoplus_{g\in G} K_i(R) \arrow[r] & \pi_{i+1}\left(BG\wedge\mathbb{K}_R^{\mathrm{GW}}\right) \arrow[r] & K_{i+1}\left(R[G]\right),
\end{tikzcd} \tag{3.3.17}$$

where the bottom composition sends the element $[f]\in K_i(R)$ in the summand in $\bigoplus_{g\in G} K_i(R)$ labelled by $g\in G$ to the element

$$[f]\star\{\mu_{g^{-1}}\}\in K_{i+1}\left(R[G]\right),$$

for which $\mu_{g^{-1}}\in\mathrm{Aut}\left(R[G]\right)$ is multiplication by g^{-1}.

Proof. Fix $g\in G$, and consider the commutative diagram

$$\begin{tikzcd}
\pi_{i+1}\left(\mathbb{S}^1\wedge\mathbb{K}_R^{\mathrm{GW}}\right) \arrow[r, "(\varphi_g)_*"] \arrow[d, "(\mathfrak{i}_t)_*"', "\cong"] & \pi_{i+1}\left(\left(\bigvee_{g\in G}\mathbb{S}^1\right)\wedge\mathbb{K}_R^{\mathrm{GW}}\right) \arrow[d, "(\mathfrak{i}_G)_*"] \\
\pi_{i+1}\left(B\langle t\rangle\wedge\mathbb{K}_R^{\mathrm{GW}}\right) \arrow[r, "(\varphi_g)_*"'] \arrow[d, "\alpha_{\mathrm{Loday}}"'] & \pi_{i+1}\left(BG\wedge\mathbb{K}_R^{\mathrm{GW}}\right) \arrow[d, "\alpha_{\mathrm{Loday}}"] \\
K_{i+1}\left(R[t^{\pm1}]\right) \arrow[r, "(\varphi_g)_*"'] & K_{i+1}\left(R[G]\right).
\end{tikzcd} \tag{3.3.18}$$

The top square commutes by chasing the definitions of the arrows. The bottom square commutes by the naturality of α_{Loday} via the group homomorphism

$$\varphi_g:\langle t\rangle\to G.$$

Now, Proposition 3.6 says the left vertical composition in Diagram (3.3.18) sends $[f]\in K_i(R)$ to

$$[f]\star\{\mu_{t^{-1}}\}\in K_{i+1}\left(R[t^{\pm1}]\right).$$

So the commutativity of Diagram (3.3.18) says the right vertical composition sends $[f]\in K_i(R)$ in the summand $\bigoplus_{g\in G} K_i(R)$ indexed by $g\in G$ to the element

$$[f]\star\{\mu_{g^{-1}}\}\in K_{i+1}\left(R[G]\right).$$

$\square$

Theorem 3.8 (An Explicit Formula for the Loday Assembly)
The filler in Diagram (3.3.6) is induced by the map

$$\begin{aligned} G \times K_i(R) &\to K_{i+1}(R[G]) \\ (g, [f]) &\mapsto [f] \star \{\mu_{g^{-1}}\}. \end{aligned}$$

Proof. Follows immediately from Proposition 3.7. □

Corollary 3.9 (The Loday Assembly on π_1, [LR05, page 708])
For a regular ring R, the Loday assembly on π_1 is given by

$$K_1(i) \oplus \Phi_1 : K_1(R) \oplus [G_{ab} \otimes K_0(R)] \to K_1(R[G]), \tag{3.3.19}$$

for which $i : R \to R[G]$ is the inclusion, and Φ_1 is induced by the map

$$\begin{aligned} G \times K_0(R) &\to K_1(R[G]) \\ (g, [P]) &\mapsto \{\mathfrak{h}_g\}, \end{aligned} \tag{3.3.20}$$

for which $\mathfrak{h}_g : P \otimes R[G] \to R \otimes R[G]$ is the automorphism given by

$$\mathfrak{h}_g(x \otimes u) := x \otimes ug^{-1}.$$

Proof. If $g \in G$, and $[P] \in K_0(R)$, then Theorem 3.8 says the Loday assembly sends $(g, [P])$ to

$$\begin{aligned} [P] \star' \{\mu_g\} &= [P] \# \{\mu_{g^{-1}}\} && \text{(Theorem 2.9)} \\ &= \{\mathfrak{h}_g\}. \end{aligned}$$

□

4. Applications to the Second K-groups of Integral Group Rings

The following SES

$$0 \to K_2(\mathbb{Z}) \oplus [G_{ab} \otimes K_1(\mathbb{Z})] \to \pi_2\left(BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\mathrm{GW}}\right) \to H_2(BG; K_0(\mathbb{Z})) \to 0 \tag{4.0.1}$$

is the extension problem of the AHSS for computing the second homotopy group $\pi_2\left(BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\mathrm{GW}}\right)$. See [Leh18, Theorem 12.2 on page 29]. Together with Theorem 3.8, we give some injectivity results for the Loday assembly on the second K-group. We begin by the following corollary:

Corollary 4.1 (The Loday Assembly on π_2 of an Integral Group Ring)
Let G be a group. The Loday assembly map

$$\alpha_{\mathrm{Loday}} : BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\mathrm{GW}} \to \mathbb{K}_{\mathbb{Z}[G]}^{\mathrm{GW}} \tag{4.0.2}$$

for the integral group ring $\mathbb{Z}[G]$ on π_2, when restricted onto the subgroup

$$K_2(\mathbb{Z}) \oplus [G_{ab} \oplus K_1(\mathbb{Z})],$$

is given by the formula:

$$K_2(i) \oplus \Phi_2 : K_2(\mathbb{Z}) \oplus [G_{ab} \oplus K_1(\mathbb{Z})] \to K_2(\mathbb{Z}[G]), \tag{4.0.3}$$

for which $i : \mathbb{Z} \to \mathbb{Z}[G]$ is the inclusion, and Φ_2 is induced by the map

$$\begin{aligned} G \times K_1(\mathbb{Z}) &\to K_2(\mathbb{Z}[G]) \\ (g, \pm 1) &\mapsto \{\pm 1, g\}_{\mathrm{St}}. \end{aligned} \tag{4.0.4}$$

In particular, the Steinberg symbol $\{1, g\}_{\mathrm{St}}$ is always the identity element in $K_2(\mathbb{Z}[G])$ for all $g \in G$.

Proof. Only the description of Φ_2 needs to be justified. Theorem 3.8 says Φ_2 is induced by the map

$$\begin{aligned} G \times K_1(\mathbb{Z}) &\to K_2(\mathbb{Z}[G]) \\ (g, \pm 1) &\mapsto \{\pm 1\} \star \{\mu_{g^{-1}}\}, \end{aligned} \tag{4.0.5}$$

and we have

$$\begin{aligned} \{\pm 1\} \star \{\mu_{g^{-1}}\} &= -\{\pm 1, \mu_{g^{-1}}\}_{\mathrm{St}} && \text{(Proposition 2.4)} \\ &= -\left\{\pm 1, g^{-1}\right\}_{\mathrm{St}} && \text{(By definition of } \mu_{g^{-1}}) \\ &= \left\{\pm 1, \left(g^{-1}\right)^{-1}\right\}_{\mathrm{St}} && \text{(Bilinearity of the Steinberg symbol)} \\ &= \{\pm 1, g\}_{\mathrm{St}} \end{aligned}$$

as desired. □

We now see that the Steinberg symbol $\{-1, g\}_{\mathrm{St}}$ is closely related to the non-injectivity problem of α_{Loday}.

Example 4.2 (Injectivity on $K_2(\mathbb{Z}[C_2])$). Denote by C_2 the cyclic group of order two, with generator g. Dunwoody showed, in [Dun75, Theorem on page 482], that the second K-group of $\mathbb{Z}[C_2]$ is generated by two Steinberg symbols:

$$\begin{aligned} K_2(\mathbb{Z}[C_2]) &= \langle \{-1, -1\}_{\mathrm{St}}, \{-1, g\}_{\mathrm{St}} \rangle \\ &\cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}. \end{aligned} \tag{4.0.6}$$

Theorem 3.8 then says the Loday assembly

$$\alpha_{\mathrm{Loday}} : (BC_2)_+ \wedge \mathbb{K}^{\mathrm{GW}}_{\mathbb{Z}} \to \mathbb{K}^{\mathrm{GW}}_{\mathbb{Z}[C_2]} \tag{4.0.7}$$

for the cyclic group of order 2 is injective on π_2. In fact, the vanishing of the second Whitehead group $\mathrm{Wh}_2(C_2)$ [Dun75, Theorem on page 482], we know it is, in fact, bijective on π_2.

In fact, we have an expression for the Steinberg symbol $\{-1,g\}_{\mathrm{St}}$ in terms of generators of the Steinberg group $\mathrm{St}(\mathbb{Z}[G])$.

Theorem 4.3
Let G be a group. The Steinberg symbol $\{-1,g\}_{\mathrm{St}} \in K_2(\mathbb{Z}[G])$ for $g \in G$ is given by

$$\{-1,g\}_{\mathrm{St}} = w_{12}(-1)\ w_{12}(-1)\ w_{12}(g)\ w_{12}(g), \tag{4.0.8}$$

where

$$w_{ij}(u) := x_{ij}(u)\ x_{ji}(-u^{-1})\ x_{ij}(u), \tag{4.0.9}$$

and the $x_{ij}(u)$s' are the generators of the Steinberg group $\mathrm{St}(\mathbb{Z}[G])$.

Proof. The proof involves playing with the presentation of the Steinberg group.

We recall from [Ros95, Lemma 4.2.15 on page 195] the identities

$$w_{ij}(u)^{-1} = w_{ij}(-u), \tag{4.0.10}$$

$$w_{ij}(u) = w_{ji}(-u^{-1}), \tag{4.0.11}$$

$$w_{k\ell}(u)\ w_{ij}(v)\ w_{k\ell}(u)^{-1} = \begin{cases} w_{ij}(v) & \text{if } i,\ j,\ k,\ \ell \text{ are all distinct,} \\ w_{\ell j}(-u^{-1}v) & \text{if } k=i, \text{ and } i,\ j,\ \ell \text{ are all distinct,} \\ w_{i\ell}(-vu) & \text{if } k=j, \text{ and } i,\ j,\ k \text{ are all distinct,} \\ w_{ji}(-u^{-1}vu^{-1}) & \text{if } k=i, \text{ and } j=\ell. \end{cases} \tag{4.0.12}$$

In particular, Equation (4.0.12) gives

$$w_{13}(u)\ w_{12}(v) = w_{23}(v^{-1}u)\ w_{13}(u), \tag{4.0.13}$$

$$w_{13}(u)\ w_{23}(v) = w_{12}(-uv^{-1})\ w_{13}(u). \tag{4.0.14}$$

Put

$$h_{ij}(u) := w_{ij}(u)\ w_{ij}(-1), \tag{4.0.15}$$

then a long and boring computation shows

$$\begin{aligned}
\{-1,g\}_{\mathrm{St}} &:= [h_{12}(-1), h_{13}(g)] \\
&= h_{12}(-1)\ h_{13}(g)\ h_{12}(-1)^{-1}\ h_{13}(g)^{-1} \\
&= w_{12}(-1)\ w_{12}(-1)\ w_{13}(g)\ w_{13}(-1)\ w_{12}(-1)^{-1}\ w_{12}(-1)^{-1}\ w_{13}(-1)^{-1}\ w_{13}(g)^{-1} \\
&= w_{12}(-1)\ w_{12}(-1)\ w_{13}(g)\ w_{13}(-1)\ w_{12}(1)\ w_{12}(1)\ w_{13}(1)\ w_{13}(-g) \\
&\quad \text{(Equation (4.0.10))}
\end{aligned}$$

$$= w_{12}(-1)\ w_{12}(-1)\ w_{13}(g)\ \underbrace{w_{13}(-1)\ w_{12}(1)}_{w_{23}(-1)\ w_{13}(-1)}\ w_{12}(1)\ w_{13}(1)\ w_{13}(-g)$$

(Equation (4.0.13))

$$= w_{12}(-1)\ w_{12}(-1)\ w_{13}(g)\ w_{23}(-1)\ \underbrace{w_{13}(-1)\ w_{12}(1)}_{w_{23}(-1)\ w_{13}(-1)}\ w_{13}(1)\ w_{13}(-g)$$

(Equation (4.0.13))

$$= w_{12}(-1)\ w_{12}(-1)\ w_{13}(g)\ w_{23}(-1)\ w_{23}(-1)\ \underbrace{w_{13}(-1)\ w_{13}(1)}_{1}\ w_{13}(-g)$$

(Equation (4.0.10))

$$= w_{12}(-1)\ w_{12}(-1)\ \underbrace{w_{13}(g)\ w_{23}(-1)}_{w_{12}(g)\ w_{13}(g)}\ w_{23}(-1)\ w_{13}(-g)$$

(Equation (4.0.14))

$$= w_{12}(-1)\ w_{12}(-1)\ w_{12}(g)\ \underbrace{w_{13}(g)\ w_{23}(-1)\ w_{13}(-g)}_{w_{31}(-g^{-1})\ w_{23}(-1)\ w_{31}(g^{-1})}$$

(Equation (4.0.11))

$$= w_{12}(-1)\ w_{12}(-1)\ w_{12}(g)\ \underbrace{w_{31}(-g^{-1})\ w_{23}(-1)\ w_{31}(g^{-1})}_{w_{21}(-g^{-1})}$$

(Equation (4.0.12))

$$= w_{12}(-1)\ w_{12}(-1)\ w_{12}(g)\ \underbrace{w_{21}(-g^{-1})}_{w_{12}(g)}$$

(Equation (4.0.11))

$$= w_{12}(-1)\ w_{12}(-1)\ w_{12}(g)\ w_{12}(g)$$

as desired. □

Theorem 4.3 motivates the following definition:

Definition 4.4 (The group $W_{12}(G)$)
Let G be a group. The subgroup $W_{12}(G)$ of $K_2(\mathbb{Z}[G])$ is generated by the Steinberg symbols:

$$W_{12}(G) := \langle \{-1, g\}_{\mathrm{St}} \mid g \in G \rangle\,. \tag{4.0.16}$$

Notice the following:

(1) The image of the map

$$\Phi_2 : G \times K_1(\mathbb{Z}) \to K_2(\mathbb{Z}[G])$$

in Equation (4.0.3) is precisely $W_{12}(G)$.

(2) Readers who are familiar with the classical definition

$$\mathrm{Wh}_2\,(G) := \frac{K_2(\mathbb{Z}[G])}{K_2(\mathbb{Z}[G]) \cap W(G)} \tag{4.0.17}$$

of the second Whitehead group proposed by Hatcher-Wagoner in [HW73, page 10] will see our group $W_{12}(G)$ follows their spirit. The group $W(G)$ is generated by the elements $w_{ij}(\pm g)$ for all $g \in G$, and for all i, j. Together with Theorem 4.3, we see that $W_{12}(G)$ is a subgroup of $W(G)$.

(3) From the functoriality of the Steinberg group, we get a functor

$$W_{12}(-) : \mathcal{G}\mathrm{roups} \to \mathcal{A}\mathrm{belian}\,. \tag{4.0.18}$$

Moreover, for every group G, there is a natural group homomorphism

$$\begin{aligned} \Psi_G : G &\to W_{12}(G) \\ g &\mapsto \{-1, g\}_{\mathrm{St}}\,, \end{aligned} \tag{4.0.19}$$

or equivalently the restriction

$$\Psi_G = \Phi_2|_{\{-1\}\times G} \tag{4.0.20}$$

for Φ_2 as in Equation (4.0.3). Non-injectivity of Ψ_G will imply the non-injectivity of the Loday assembly on π_2.

Example 4.5 (Injectivity on $K_2\left(\mathbb{Z}\left[C_2^{\oplus n}\right]\right)$). This example is due to Daniel Ramras.

Denote by C_2 the cyclic group of order 2. Fix $n \in \mathbb{N}$, and consider the n-fold direct sum. If $x \in C_2^{\oplus n}$ is non-zero, then there exists $1 \leq i \leq n$ such that the image $\mathrm{proj}_i(x)$ of x under the i-th projection map

$$\mathrm{proj}_i : C_2^{\oplus n} \to C_2$$

is non-zero. The functoriality of $W_{12}(-)$ then gives a commutative diagram

$$\begin{tikzcd} C_2^{\oplus n} \arrow[r, "\Psi_{C_2^{\oplus n}}"] \arrow[d, "\mathrm{proj}_i"'] & W_{12}(C_2^{\oplus n}) \arrow[d, "W_{12}(\mathrm{proj}_i)"] \\ C_2 \arrow[r, "\Psi_{C_2}"'] & W_{12}(C_2). \end{tikzcd} \tag{4.0.21}$$

We saw in Example 4.2 that Ψ_{C_2} is injective, so the commutativity of Diagram (4.0.21) says $\Psi_{C_2^{\oplus n}}(x) \in W_{12}(C_2^{\oplus n})$ is non-zero.

Let us recap what we did here. Fix a non-zero element $x \in C_2^{\oplus n}$. There exists $1 \leq i \leq n$ such that $\mathrm{proj}_i(x) \in C_2$ is non-zero. Because Ψ_{C_2} is injective, we must have

$$\left(W_{12}(\mathrm{proj}_i) \circ \Psi_{C_2^{\oplus n}}\right)(x) \neq 0,$$

and consequently $\Psi_{C_2^{\oplus n}}(x) \neq 0$, proving the group homomorphism

$$\Psi_{C_2^{\oplus n}} : C_2^{\oplus n} \to W_{12}(C_2^{\oplus n})$$

is injective for any $n \in \mathbb{N}$.

As a result, the Loday assembly map

$$\alpha_{\text{Loday}} : B\left(C_2^{\oplus n}\right)_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\text{GW}} \to \mathbb{K}_{\mathbb{Z}\left[C_2^{\oplus n}\right]}^{\text{GW}}$$

for the group $C_2^{\oplus n}$ is injective on π_2 when restricted onto the subgroup $K_2(\mathbb{Z}) \oplus \left[C_2^{\oplus n} \otimes K_1(\mathbb{Z})\right]$.

Example 4.6 (Limitation). Our formula does not work well on studying the K-theory of the integral group ring $\mathbb{Z}[C_p]$ of cyclic group C_p of prime order $p > 2$. Since $C_p \otimes C_2 = 0$, the map Φ_2 in Equation (4.0.4) is the zero map. Because $H_2\left(BC_p; \mathbb{Z}\right) = 0$ for all p, the SES in Equation (4.0.1) gives

$$\pi_2\left(BC_{p+} \wedge \mathbb{K}_{\mathbb{Z}}^{\text{GW}}\right) \cong K_2(\mathbb{Z}), \tag{4.0.22}$$

and therefore we learn nothing about structures of $K_2(\mathbb{Z}[C_p])$ from the assembly map approach, other than it contains $K_2(\mathbb{Z})$ as a subgroup.

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